

KYUSYUR, Russian Federation

WMO: 219210

Lat: 70.683N

Lon: 127.400E

Elev: 108

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-57.8	-55.3	-60.8	0.1	-56.9	-58.3	0.2	-54.9	29.4	-12.1	25.7	-13.5	1.5	50	0.919

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.6	76.7	61.3	72.2	59.7	67.7	57.5	63.7	72.8	61.2	70.2	58.5	66.5	8.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.9	77.4	67.6	56.9	69.2	65.8	54.0	62.4	63.3	29.2	73.5	27.4	70.4	25.5	66.7	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.5	21.8	18.9	-60.7	84.3	3.3	3.8	-63.1	87.0	-65.1	89.2	-67.0	91.3	-69.4	94.0		
(5)			-61.0	67.4	3.5	2.2	-63.5	68.9	-65.6	70.2	-67.5	71.5	-70.0	73.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	-32.9	-30.7	-13.2	9.0	28.3	49.8	54.4	50.9	37.8	13.4	-17.8	-31.7		
	DBStd	34.95	13.92	14.72	15.98	12.15	10.43	9.99	8.43	6.55	6.84	12.56	15.09	14.64		
	HDD50	15003	2570	2259	1960	1230	674	124	45	68	370	1136	2034	2533		
	HDD65	20110	3035	2679	2425	1680	1137	469	346	439	817	1601	2484	2998		
	CDD50	401	0	0	0	0	2	117	182	96	4	0	0	0		
	CDD65	31	0	0	0	0	0	12	18	1	0	0	0	0		
	CDH74	248	0	0	0	0	0	98	145	5	0	0	0	0		
	CDH80	43	0	0	0	0	0	15	28	0	0	0	0	0		
	WSAvg	6.9	4.7	5.1	6.3	8.4	9.3	9.2	8.7	7.9	7.3	7.3	4.7	4.3		
	PrecAvg	13.5	0.6	0.6	0.7	0.6	0.7	1.4	2.0	2.1	1.6	1.6	0.9	0.7		
	PrecMax	18.0	1.7	1.6	2.2	2.3	2.0	2.9	6.3	5.7	3.6	3.1	2.1	1.6		
(15) Precipitation	PrecMin	8.5	0.0	0.0	0.1	0.0	0.0	0.3	0.4	0.5	0.3	0.5	0.1	0.2		
	PrecStd	2.6	0.4	0.4	0.5	0.5	0.5	0.7	1.2	1.1	0.9	0.7	0.5	0.4		
	DB	6.1	11.8	29.7	42.5	61.7	81.9	83.5	74.3	62.2	35.6	23.8	6.2			
	MCWB	5.1	10.4	27.3	36.0	48.6	62.4	65.2	61.3	51.8	33.2	22.5	5.2			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	-2.4	1.4	23.5	36.9	52.5	76.6	78.1	70.2	54.9	32.6	15.4	0.5			
	2% MCWB	-3.0	0.5	21.9	31.9	43.3	59.7	63.5	59.1	47.4	30.7	14.4	-0.3			
	5% DB	-7.2	-4.1	18.0	32.7	47.2	71.8	73.3	66.4	50.7	30.8	9.4	-4.4			
	5% MCWB	-7.7	-4.8	16.2	28.8	40.2	57.7	61.9	57.3	45.3	28.8	8.6	-5.1			
	10% DB	-12.0	-8.4	11.8	28.2	42.9	66.6	68.5	62.2	47.6	28.6	3.7	-9.5			
	10% MCWB	-12.4	-9.0	10.4	24.8	37.3	55.3	59.0	55.5	43.2	26.8	3.0	-10.0			
	0.4% WB	5.2	10.5	27.4	35.9	49.1	64.2	67.4	64.3	53.2	33.5	22.6	5.3			
	0.4% MCDB	5.9	11.9	29.6	41.8	60.9	76.5	79.0	70.0	59.6	35.0	23.5	5.8			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	-3.0	0.6	22.3	32.3	43.7	61.3	64.9	60.9	48.9	31.2	14.5	-0.3			
	2% MCDB	-2.4	1.3	23.6	36.3	51.2	73.7	74.6	67.5	53.3	32.6	15.5	0.3			
	5% WB	-7.7	-4.7	16.2	28.9	40.4	58.8	62.9	58.4	46.0	29.0	8.6	-5.0			
	5% MCDB	-7.2	-4.1	17.9	32.6	46.8	69.5	71.5	64.6	50.1	30.7	9.3	-4.3			
	10% WB	-12.5	-9.0	10.4	24.8	37.6	55.6	60.1	56.1	43.5	26.8	3.1	-10.0			
	10% MCDB	-12.1	-8.4	11.7	28.0	42.7	66.2	68.2	61.5	47.1	28.6	3.8	-9.5			
	MDBR	12.5	12.1	15.7	17.9	12.4	14.8	14.6	13.5	10.3	9.9	12.6	11.9			
	5% DB	16.8	16.4	15.5	19.8	17.5	22.0	21.0	20.9	16.4	7.9	16.0	17.8			
(36) Mean Daily Temperature Range	MCWBR	16.2	15.8	14.0	16.1	10.6	10.5	11.0	12.1	10.4	7.1	15.4	17.0			
	MCDBR	16.2	15.6	15.2	19.4	16.1	20.2	19.0	17.7	14.6	8.1	16.1	17.0			
	MCWBR	15.8	15.2	14.1	16.0	10.2	11.2	11.2	11.5	9.8	7.5	15.5	16.7			
	MCDBR	15.8	15.2	14.1	16.0	10.2	11.2	11.2	11.5	9.8	7.5	15.5	16.7			
(40) Clear-Sky Solar Irradiance	taub	N/A	0.210	0.285	0.316	0.349	0.351	0.415	0.395	0.299	0.219	N/A	N/A			
	taud	N/A	1.988	2.033	2.066	2.153	2.411	2.236	2.245	2.419	2.033	N/A	N/A			
	Ebn at Noon	0	154	217	252	259	266	238	226	225	141	0	0			
	Edh at Noon	0	12	26	36	37	30	35	30	18	11	0	0			
(44) All-Sky Solar Radiation	RadAvg	0	102	593	1374	1855	1778	1570	1040	505	141	0	0			
	RadStd	0	11	42	65	116	171	110	134	67	22	0	0			

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+1.89	N/A	N/A	N/A	N/A	N/A	-743	-709	N/A	N/A

Nomenclature: See separate page